NLP Project

Mid-Project Submission

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Problem Statement

Digital health platforms and medical online forums have dramatically changed patient engagement in healthcare: every day, millions of people consult the web for health related questions. However, the number of medical and health related textual sources on the web have led to a huge amount of unstructured and heterogeneous data. A difficult issue in building automated Consumer Health Question Answering systems is the nature of the Consumer Health Questions (CHQs); questions can be lengthy, unstructured, and sometimes contain tangential and redundant information. Patients tend to nest their real health concerns within detailed personal narratives filled with details about emotional states, family history, location, or other irrelevant facts. The noise, along with the resulting semantic gap between lay language and medical terms, exacerbates the problem in subsequent natural language processing tasks.

CHQ summarization attempts to tackle this problem by distilling long consumer health questions into succinct, diagnostically-relevant medical questions while retaining all essential diagnostic aspects. As a preprocessing step, abstractive summarization could enable greater efficiency, higher accuracy, and improved retrieval for subsequent medical QA systems and information retrieval pipelines. We believe that a shorter and clinically specific query will achieve better answers from reputable medical knowledge sources.

The low-resource nature of the medical domain, however, poses great difficulties for building dependable summarization systems. State-of-the-art language generation systems demand substantial amounts of labeled training data, but, the creation of medical datasets is expensive, time consuming, and further restricted by patient privacy. Medical annotations also demand specialized expertise to make sure important diagnostic features are maintained and not spurious information is included. For this reason, instead, state-of-the-art models are commonly finetuned with a small amount of data, sometimes as little as under 1000 sentences, which results in issues such as over-fitting, catastrophic forgetting, semantic shift, and clinical hallucination.

However, if an accurate CHQ summarization system is created, it would have great impact in several medical applications such as EHR structuring, clinical decision support, medical chatbots, and even for automated medical triaging systems, enabling improved healthcare efficiency and scalability in resource-constrained environments.

Literature survey

Recent research on healthcare question summarization shows a clear shift from traditional Seq2Seq and LSTM-based models to Transformer-based architectures such as **BART, T5, PEGASUS, and ProphetNet**. Earlier neural summarization methods established the feasibility of abstractive summarization but struggled with long-range dependencies and limited semantic understanding. Transformer models significantly improved performance by leveraging self-attention and large-scale pretraining. Among them, **BART** is especially effective for summarization because its denoising pretraining aligns well with the task of removing irrelevant details from verbose patient queries, while **T5** provides a flexible text-to-text framework with strong factual faithfulness. **PEGASUS** is particularly promising in low-resource settings due to its summary-oriented pretraining objective.

In the healthcare domain, summarization is more challenging because consumer health questions are often lengthy, informal, and filled with non-essential information, while datasets such as **MEQSUM** remain very small. This low-resource nature makes models prone to **overfitting, semantic drift, and clinical hallucination**. To address data scarcity, studies increasingly explore **data augmentation**, especially **Round-Trip Translation (RTT)**, combined with filtering strategies such as **FQD, PRQD, and QSV** to preserve semantic quality while improving diversity. Overall, existing work suggests that while pretrained Transformers achieve strong ROUGE scores, ensuring **clinical accuracy and factual consistency** remains a major research challenge.

Description Of Dataset

The MEQSUM dataset has been used for training and evaluation machine learning. The dataset has been taken from US National Library of Medicine,National institute of health .The dataset contains 1000 pairs of question-summary . The structural schema of the dataset consists of three primary string type datafields: 1) CHQ , 2) Summary , 3) File.

Example Input/Output Pairs

Raw Consumer Health Question (CHQ)	Expert Annotated Target (Summary)
"SUBJECT: who and where to get cetirizine - D MESSAGE: I need/want to know who manufacturer Cetirizine. My Walmart is looking for a new supply and are not getting the recent"	Who manufactures cetirizine?
"Williams' syndrome I would like to have my daughter tested for William's syndrome. Could you please tell me where I would go or who does it in my area? Thank you!!"	Where can I get genetic testing for william's syndrome?
"ClinicalTrials.gov - Question - general information. Both my parents died in TX with multiple myeloma... I was unaware of a genetic test for MM I am 63 yr old female, in good health, and where could I get the genetic test for MM and approx cost for testing?"	Where can I get genetic testing for multiple myeloma, and what is the cost?
"MESSAGE: Will storing my medication in the Do freezer reduce their efficacy? I have many and more related questions about storing pills at various temperatures... The drugs that I am concerned about are digoxin and lisinopril. "	Will storing digoxin and lisinopril in the Do freezer reduce their efficacy?

Baseline Results

The baseline have been compared on the metrics of Rouge scores.The following score have been used:

1) ROUGE-1 (Unigrams) : Measures single word overlap
2) ROUGE-2 (Bigrams) : Measures 2-word sequence overlap
3) ROUGE-L (Longest common Subsequence) : It looks for the longest matching sequence of words in order

Baseline 1 : ProphetNet

ProphetNet is a pre-trained encoder-decoder transformer model designed for sequence-to-sequence tasks like summarization, translation, and question answering. ProphetNet uses "future n-gram prediction" instead of standard next-token prediction.So it is autoregressive in nature.

The model was trained on 3 experimental variations:
1) Original Dataset
2) Original Dataset+RTT
3) Original Dataset + RTT + Filtering

Metrics	**Original Dataset**	**Original Dataset+RTT**	**Original Dataset + RTT + Filtering**
ROUGE-1	0.547039252560069	0.5429611987194627	0.8919593691825867,
ROUGE-2	0.3437291799845507	0.3617281135250685	0.16476624193847422
ROUGE-L	0.5040162121594733	0.5075197676124837	0.002105263157894737
Evaluation Loss	0.4045373499393463	0.4531271457672119	0.16338528955752185

Baseline 2: BART

BART (Bidirectional and Auto-Regressive Transformers) is a pre-trained encoder-decoder transformer model designed for sequence-to-sequence tasks like summarization, translation, and text generation. It is pre-trained by corrupting the input text with arbitrary noise functions (such as token masking, text infilling and sentence permutation) and learning to reconstruct the original, uncorrupted text. Since its decoder predicts the original tokens one by one, it is also autoregressive in nature.

The model was trained on 3 experimental variations:

1) Original Dataset
2) Original Dataset+RTT
3) Original Dataset + RTT + Filtering

Metrics	Original Dataset	Original Dataset+RTT	Original Dataset + RTT + Filtering
ROUGE-1	0.5496995136812989	0.5276257738379875	0.5644803862138834
ROUGE-2	0.3598711404742293	0.37182140640079125	0.3842464547418305
ROUGE-L	0.5190834597425165	0.5082766447363816	0.5407493541396625
Evaluation Loss	0.3496124744415283	0.3513202667236328	0.3461708724498749